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CHARACTERIZATION OF AVOCADOS CV. LAMB HASS FROM THE MEDITERRANEAN AREA IN SPAIN AND THE RELATIONSHIP AMONG QUALITY ATTRIBUTES

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OBJECTIVES

a) To evaluate the evolution of key physicochemical parameters of 'Lamb Hass' avocado during harvest and subsequent ripening throughout the season.

b) To identify possible correlations between the parameters evaluated.

MATERIALS AND METHODS

'Lamb Hass' avocado

Three Harvests: March, April, May
Selection, and random distribution of batches

Stored at 20 °C 90 % RH

Evaluation of non-destructive (**ND**) and destructive (**D**) quality parameters, every 2 days

- ***Dry matter:***
NIR (Felix; F-751) and oven methods
- ***Flesh firmness:***
Instron (Newton needed to break the flesh (D), penetrometer (Kg, D) and durofel (neck and equatorial area (ND)
- ***External color:*** CIELab coordinates (ND)

RESULTS

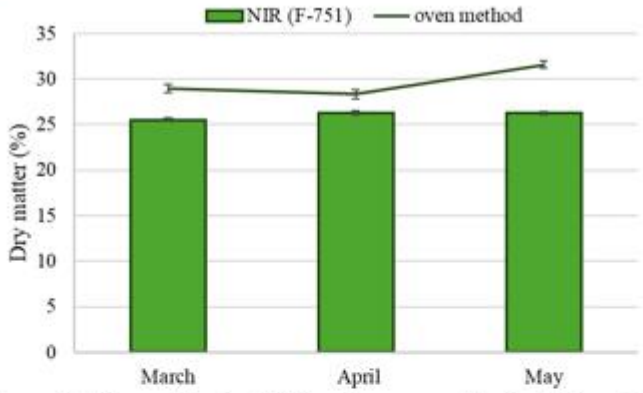


Figure 1. Dry matter by NIR and oven methods of 'Lamb Hass' avocado at harvest moment.

Figure 2. External appearance of 'Lamb Hass' avocado.

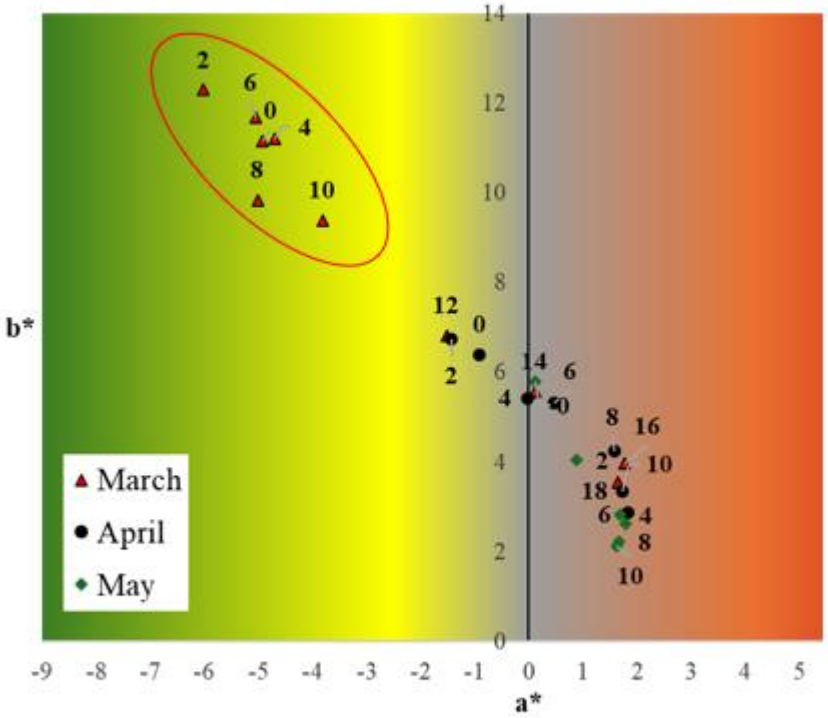
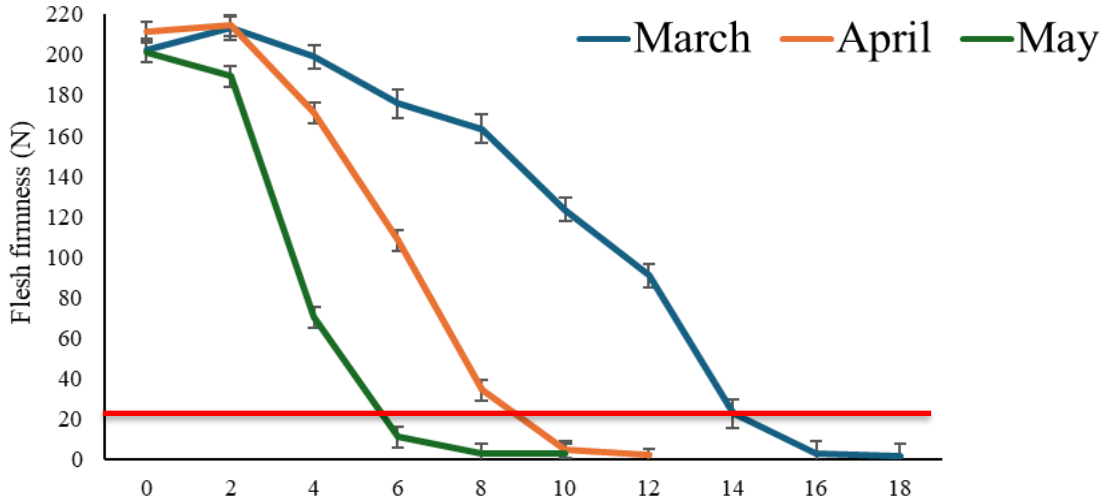
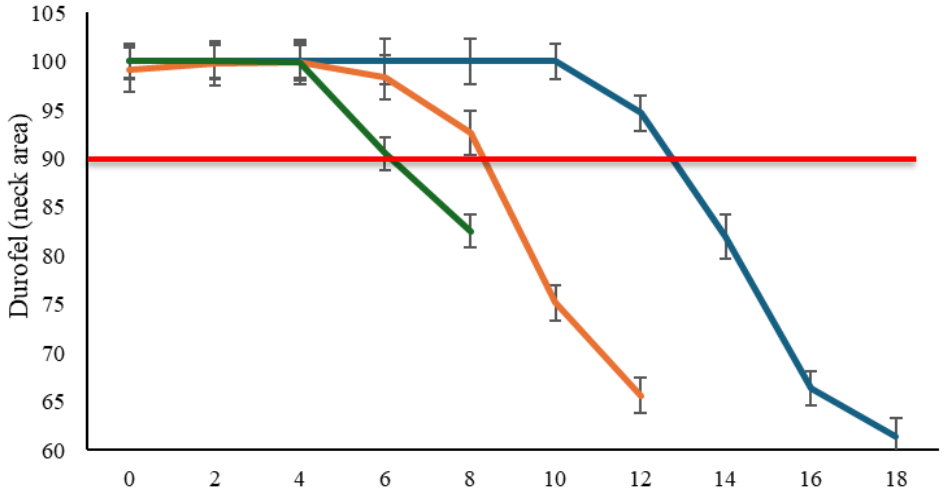
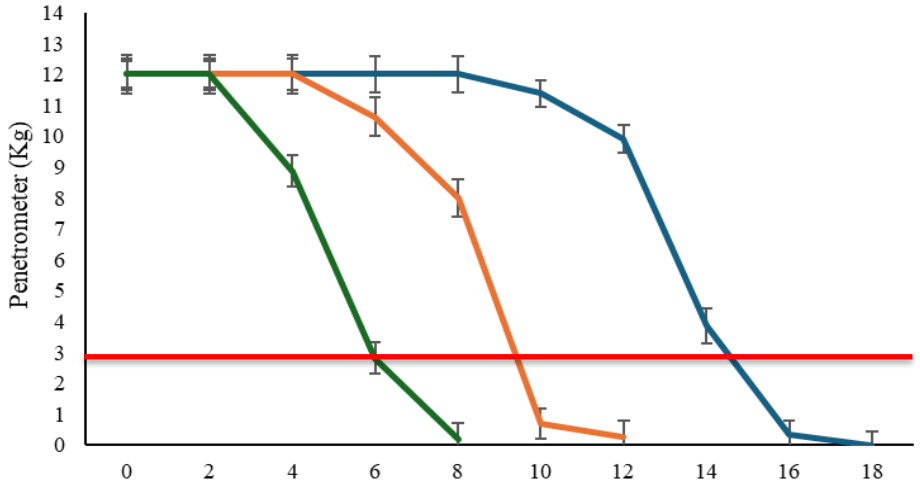


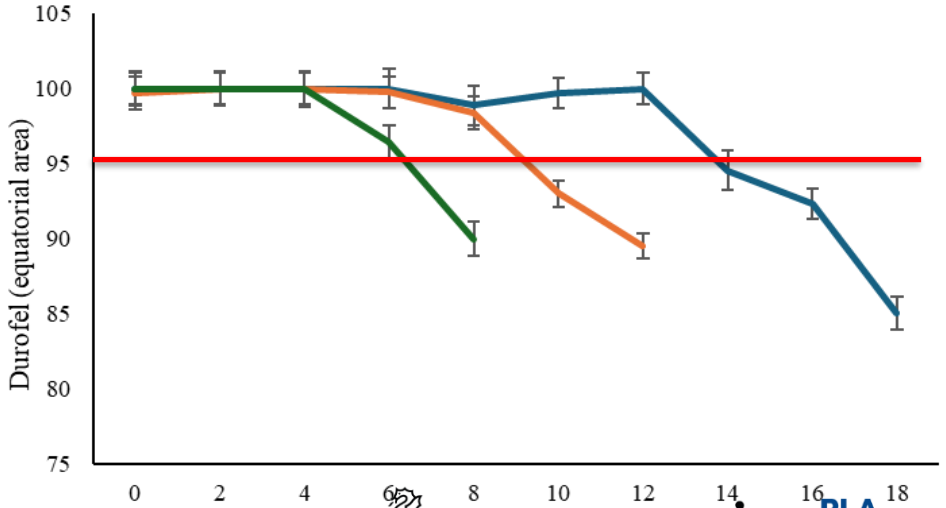
Figure 3. Evolution of coordinates (a* and b*) from skin color of 'Lamb Hass' avocado fruit during storage time at 20°C until maturation.
(the numbers in the graph represent the days of storage at 20°C for each harvest.)

RESULTS









Days at -20 °C

RESULTS

	F. Firmness (Instron)	Durofel (eq)	Durofell (neck)	Penetrom	L* skin (harvest)	a* skin (harvest)	b* skin (harvest)	L* skin (after)	a* skin (after)	b* skin (after)	Dry matter (oven)
F. Firmness (Instron)											
Durofel (equatorial)	**0.50**										
Durofell (neck)	**0.67**	**0.58**									
Penetrometer	**0.89**	**0.59**	**0.75**								
L* skin (harvest)	0.04	0.02	-0.09	0.03							
a* skin (harvest)	-0.03	0.01	0.11	-0.03	**-0.93**						
b* skin (harvest)	0.02	-0.02	-0.13	0.02	**0.92**	**-0.97**					
L* skin (after storage)	**0.61**	**0.34**	**0.49**	**0.58**	**0.52**	**-0.50**	**0.51**				
a* skin (after storage)	**-0.62**	**-0.32**	**-0.43**	**-0.57**	**-0.51**	**0.54**	**-0.53**	**-0.91**			
b* skin (after storage)	**0.66**	**0.35**	**0.42**	**0.59**	**0.56**	**-0.59**	**0.61**	**0.90**	**-0.95**		
Dry matter (oven)	0.09	0.01	0.09	0.05	**-0.46**	**0.53**	**-0.48**	**-0.25**	**0.27**	**-0.27**	
Dry matter (Felix)	-0.02	0.03	0.05	-0.04	**-0.55**	**0.62**	**-0.61**	**-0.30**	**0.27**	**-0.34**	**0.51**

Numbers in bold type represents significant correlation at $p < 0.001$. Pearson correlations between each pair of variables.

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CONCLUSIONS

- No differences in dry matter were found at harvest time, **but the time required to reach commercial firmness varied with harvest date**.
- Most fruit achieved **commercial firmness (≤ 20 N) at 14, 8, and 6 days at 20 °C for the March, April, and May harvests**, respectively.
- The **main correlations** between the characteristics were:
  - ✓ Destructive and Non-Destructive firmness parameters.
  - ✓ Skin color coordinates after storage with all parameters evaluated.
  - ✓ Dry matter with color coordinates before and during ripening.
- **This work advances to consolidate the promising Spanish Lamb Hass avocado in European markets**